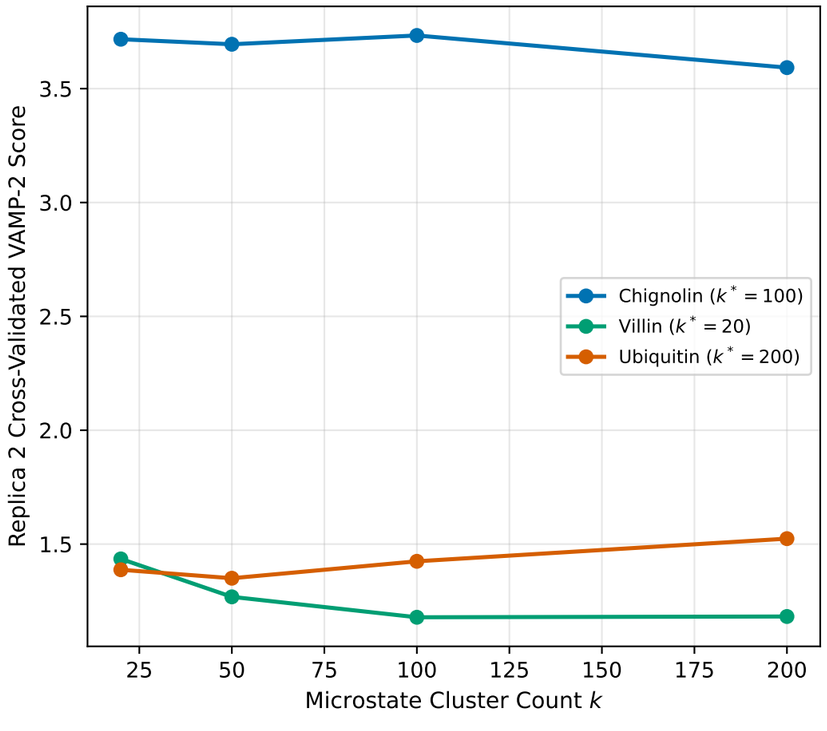
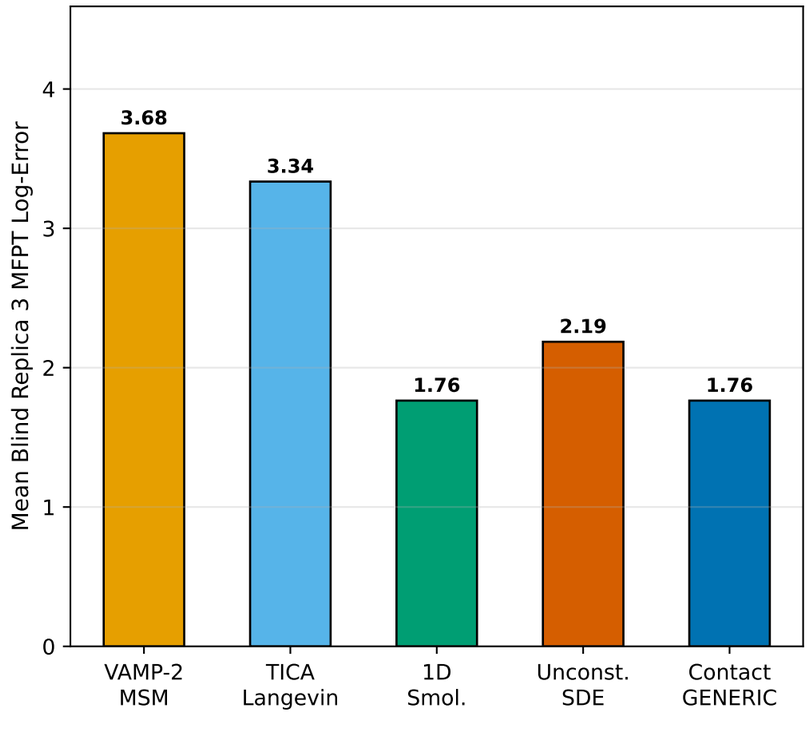
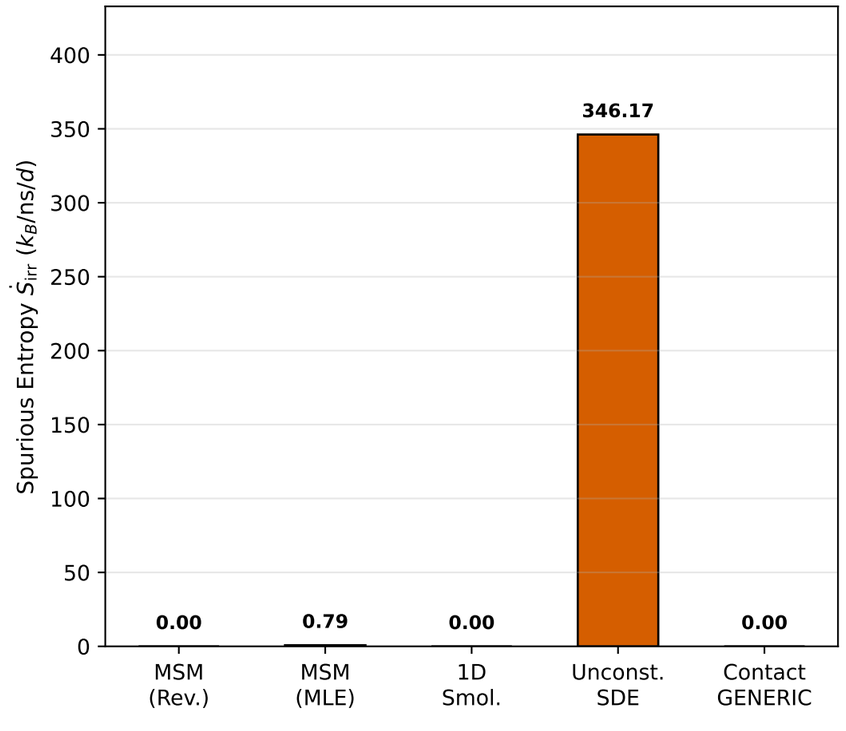
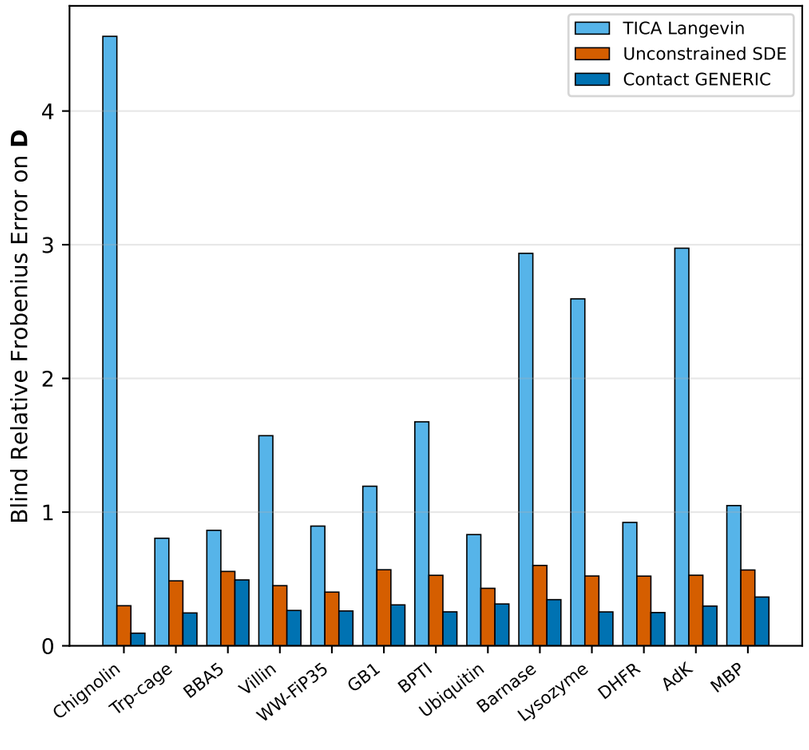
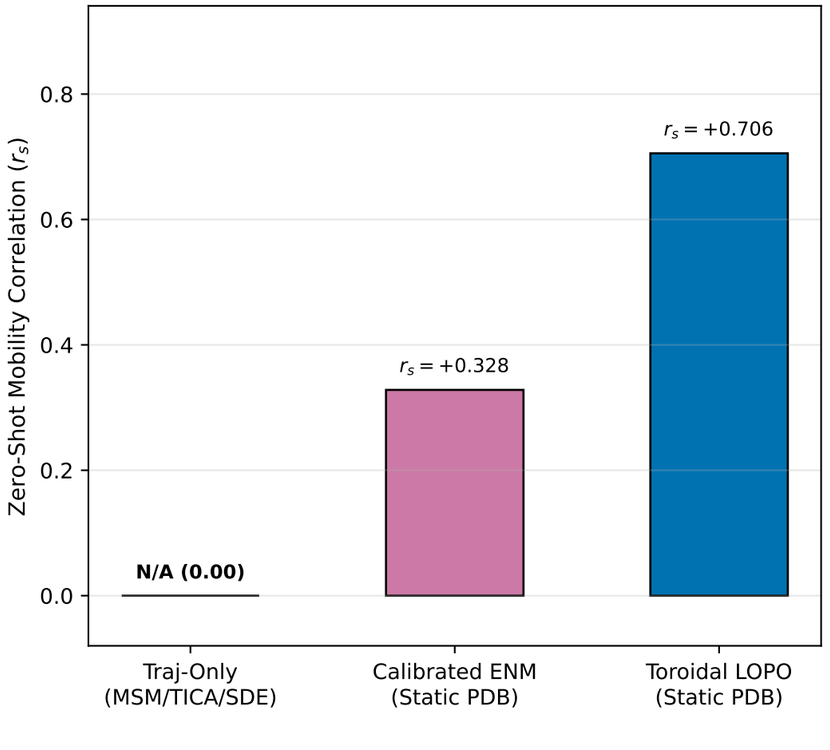
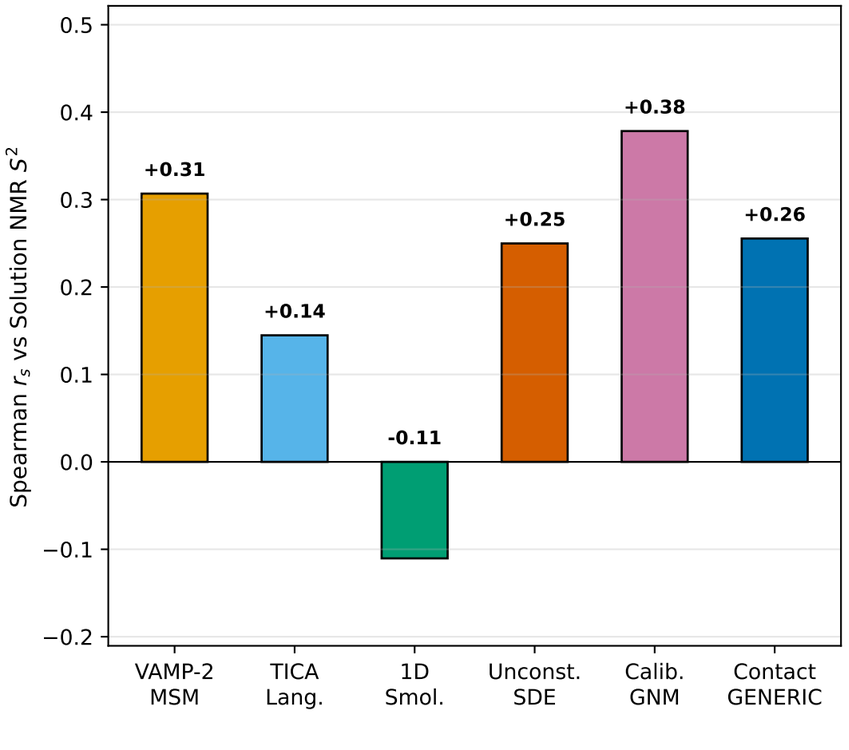

(A) Validation Tuning Curves (Replica 2)
Replica 2 Cross-Validated VAMP-2 Score
Microstate Cluster Count k
Chignolin ($k^* = 100$)
Villin ($k^* = 20$)
Ubiquitin ($k^* = 200$)


(B) Metric A — Barrier-Crossing Kinetics
Mean Blind Replica 3 MFPT Log-Error
3.68
3.34
1.76
2.19
1.76
VAMP-2 MSM
TICA Langevin
1D Smol.
Unconst. SDE
Contact GENERIC


(C) Metric B — Detailed Balance & Cycle Flux
Spurious Entropy $\dot{S}_{irr}$ (k_B/ns/d)
0.00
0.79
0.00
346.17
0.00
MSM (Rev.)
MSM (MLE)
1D Smol.
Unconst. SDE
Contact GENERIC


(D) Metric C — Operator Reconstruction ($d = 18 \rightarrow 738$)
Blind Relative Frobenius Error on D
TICA Langevin
Unconstrained SDE
Contact GENERIC
Chignolin
Trp-cage
BBA5
Villin
WW-FIP35
GB1
BPTI
Ubiquitin
Barnase
Lysozyme
DHFR
AdK
MBP


(E) Metric D — 13-Fold Zero-Shot Transfer
Zero-Shot Mobility Correlation (r_s)
N/A (0.00)
$r_s = +0.328$
$r_s = +0.706$
Traj-Only (MSM/TICA/SDE)
Calibrated ENM (Static PDB)
Toroidal LOPO (Static PDB)


(F) Metric E — Solution NMR S^2 Fidelity
Spearman r_s vs Solution NMR S^2
+0.31
+0.14
-0.11
+0.25
+0.38
+0.26
VAMP-2 MSM
TICA Lang.
1D Smol.
Unconst. SDE
Calib. GNM
Contact GENERIC